

Derivation of the Ideal Gas Law from Statistical Thermodynamics

This document derives the ideal gas law starting from quantum eigenstates of a particle in a three-dimensional box and using the canonical partition function.

1. Quantum Energy Eigenstates

For a particle confined to a cubic box of side length L , the energy eigenvalues are:

$$E = (\hbar^2 / 8mL^2) (n_x^2 + n_y^2 + n_z^2)$$

2. Single-Particle Partition Function

$$Z_1 = \sum \exp(-\beta E) \rightarrow \int \exp(-\beta p^2/2m) d^3p d^3x / h^3$$

3. Classical Limit

At high temperatures the energy levels are closely spaced and the sum becomes an integral, yielding:

$$Z_1 = V (2\pi mkT / h^2)^{3/2} = V / \lambda^3$$

4. N-Particle Partition Function

For indistinguishable particles:

$$Z_N = (1/N!) (V/\lambda^3)^N$$

5. Free Energy and Pressure

$$F = -kT \ln Z_N$$

$$P = -(\partial F / \partial V)_T = NkT / V$$

6. Ideal Gas Law

Rearranging gives the ideal gas law:

$$PV = NkT$$

This result emerges from the absence of interactions and the quadratic dependence of energy on momentum.